

The Grothendieck Criterion Extends to Bounded-Degree Type-I Marcinkiewicz Spaces

Candidate strong partial-result packet

arXiv:1912.01162, Introduction

12 August 2026

Status. Candidate strong partial answer, subject to expert review. The scalar criterion extends exactly to every fixed finite matrix amplification and, more generally, the positive result extends to all semifinite type-I von Neumann algebras of uniformly bounded degree. The source’s unrestricted noncommutative question remains open here.

1 Source question

De Pagter–Sukochev characterize the increasing concave functions ψ for which the scalar Marcinkiewicz space $M_\psi(0, \gamma)$ is Grothendieck. In their Introduction, page 2, they ask:

“It seems to be an interesting open question whether the results presented in this article also hold for noncommutative Marcinkiewicz spaces.”

Their conditions are

$$\begin{aligned} \text{(A)} \quad \gamma = \infty, \quad & \liminf_{t \downarrow 0} \frac{\psi(2t)}{\psi(t)} > 1, \quad & \liminf_{t \rightarrow \infty} \frac{\psi(2t)}{\psi(t)} > 1; \\ \text{(B)} \quad \gamma < \infty, \quad & \liminf_{t \downarrow 0} \frac{\psi(2t)}{\psi(t)} > 1. \end{aligned}$$

They prove that (A) or (B), as appropriate, is equivalent to $M_\psi(0, \gamma)$ being Grothendieck, and gives sufficiency over arbitrary σ -finite scalar measure spaces [1, Theorems 5.6 and 6.3].

article are also presented (with different proofs). It seems to be an interesting open question whether the results presented in this article also hold for noncommutative Marcinkiewicz spaces. Finally, we mention that all Banach

Figure 1: The open question in the source PDF, page 2.

2 Definitions

Let $(\mathcal{M}, \tau)$ be a semifinite von Neumann algebra with normal faithful semifinite trace, and let $\mu(x)$ denote the generalized singular-value function of a τ -measurable operator x . The noncommutative Marcinkiewicz space is

$$M_\psi(\mathcal{M}, \tau) = \{x : \mu(x) \in M_\psi(0, \tau(1))\}, \quad \|x\|_{M_\psi(\mathcal{M})} = \|\mu(x)\|_{M_\psi}.$$

We call $(\mathcal{M}, \tau)$ a *bounded-degree type-I model* if, for some $d < \infty$,

$$(\mathcal{M}, \tau) \cong \bigoplus_{r=1}^d (L_\infty(\Omega_r, \nu_r) \overline{\otimes} M_r, \int_{\Omega_r} \text{Tr}_r \, d\nu_r). \quad (1)$$

This is the standard central form of a semifinite subhomogeneous von Neumann algebra. Zero summands are allowed.

3 Proof Intuition

A fixed matrix algebra introduces only finitely many noncommutative coordinates. Compressing an operator-valued function by two matrix units extracts one scalar entry and cannot increase any symmetric operator norm. Conversely, summing the finitely many entry corners reconstructs the operator. Hence the noncommutative space is Banach-isomorphic to a finite product of scalar Marcinkiewicz spaces. Grothendieckness is stable under finite products, and a scalar corner is complemented, giving the converse in the homogeneous case.

4 The matrix-corner lemma

Lemma 1. *Let E be any fully symmetric Banach function space and $\mathcal{M} = L_\infty(\Omega) \overline{\otimes} M_r$ with the tensor trace. For $x = (x_{ij}) \in E(\mathcal{M})$,*

$$\max_{i,j} \|x_{ij}e_{ij}\|_{E(\mathcal{M})} \leq \|x\|_{E(\mathcal{M})} \leq \sum_{i,j} \|x_{ij}e_{ij}\|_{E(\mathcal{M})} \leq r^2 \max_{i,j} \|x_{ij}e_{ij}\|_{E(\mathcal{M})}. \quad (2)$$

Consequently, the entry map is a Banach isomorphism from $E(\mathcal{M})$ onto the finite product of its r^2 scalar corner spaces.

Proof. Let e_{ij} be the constant matrix units. The ideal property of a symmetric operator space gives

$$\|e_{ii}xe_{jj}\|_{E(\mathcal{M})} \leq \|e_{ii}\|_\infty \|x\|_{E(\mathcal{M})} \|e_{jj}\|_\infty \leq \|x\|_{E(\mathcal{M})}.$$

But $e_{ii}xe_{jj} = x_{ij}e_{ij}$, proving the first inequality. Since $x = \sum_{i,j} x_{ij}e_{ij}$, the triangle inequality proves the second, and the third is immediate. Every family of measurable scalar entries defines a measurable M_r -valued function; (2) shows that it lies in $E(\mathcal{M})$ exactly when all its finitely many corners lie there. Thus the entry map and its inverse are bounded. $\square$

For a corner over a measure space of mass γ_r , the singular-value function of fe_{ij} is the decreasing rearrangement of $|f|$, followed by zeros. Since ψ is increasing, the Marcinkiewicz supremum after γ_r cannot exceed its value at γ_r . Thus the corner space is precisely the scalar $M_\psi(\Omega_r, \nu_r)$ with its usual norm.

5 Partial solution

Theorem 1. *Let ψ satisfy condition (A) when $\tau(1) = \infty$ and condition (B) when $\tau(1) < \infty$. For every bounded-degree type-I model $(\mathcal{M}, \tau)$ in (1), the noncommutative Marcinkiewicz space $M_\psi(\mathcal{M}, \tau)$ is a Grothendieck Banach space.*

For the homogeneous atomless model

$$\mathcal{M} = L_\infty(0, \gamma) \overline{\otimes} M_r \quad (1 \leq r < \infty),$$

the converse also holds: $M_\psi(\mathcal{M}, \tau)$ is Grothendieck if and only if the applicable condition (A) or (B) holds. In particular, every finite matrix amplification of weak L_p , $1 < p < \infty$, is Grothendieck.

Proof. Apply Lemma 1 on each of the finitely many central summands in (1). This identifies $M_\psi(\mathcal{M}, \tau)$ with a finite product of scalar Marcinkiewicz spaces $M_\psi(\Omega_r, \nu_r)$, one for each triple (r, i, j) .

If $\tau(1) < \infty$, every $\nu_r(\Omega_r)$ is finite and (B) gives the Grothendieck property of every scalar factor by Theorem 6.3 of [1]. If $\tau(1) = \infty$, a component has either finite or infinite measure. The zero-end condition in (A) gives (B) on each finite component, while both parts of (A) apply to every infinite component. Again every scalar factor is Grothendieck. A finite product of Grothendieck spaces is Grothendieck: its dual is the corresponding finite product of the duals, so weak-star and weak convergence can be checked coordinatewise. Banach isomorphisms preserve the property, proving the first assertion.

Now take the homogeneous atomless model. Sufficiency was just proved. The map $P(x) = e_{11}xe_{11}$ is a bounded projection whose range is isometric to the scalar corner $M_\psi(0, \gamma)$. Every complemented subspace of a Grothendieck space is Grothendieck. Therefore, if the noncommutative space is Grothendieck, then so is $M_\psi(0, \gamma)$; Theorem 5.6 of [1] forces (A) or (B). Finally, for $\psi(t) = t^{1-1/p}$ both endpoint ratios are $2^{1-1/p} > 1$. $\square$

6 Verification and limitations

The accompanying script tests (2) on random finite atomic matrix fields using the discrete Marcinkiewicz norm and verifies exact reconstruction from entries. This is a sanity check; the proof is exact.

The finite-degree hypothesis is essential to the present method: the inverse entry-map bound grows as r^2 . For unbounded type-I degree or type-II algebras, the argument would require a dimension-free decomposition. The source proof also cannot simply be copied, because a noncommutative symmetric space is not a Banach lattice and lacks the suprema and dual lattice homomorphisms used in Lotz’s disjointization argument. Eight focused upgrade attempts, recorded in the run attempt note, did not remove these obstacles.

A bounded search through 12 August 2026 checked the run indexes; exact source wording and title/author searches; and combinations of “noncommutative Marcinkiewicz”, “Grothendieck property”, “symmetric operator space”, “matrix amplification”, and “subhomogeneous”. It found the source, adjacent work on noncommutative $L_p + L_q$ spaces and on weak Grothendieck compactness, but no general resolution or the finite-degree theorem above. This is not a claim of priority.

Human-review recommendation. Check that the scalar corner norm is exactly the zero-extension Marcinkiewicz norm and verify the standard central decomposition convention in (1). The matrix inequalities and Grothendieck transfer are then immediate.

References

- [1] B. de Pagter and F. A. Sukochev, *The Grothendieck property in Marcinkiewicz spaces*, Indagationes Mathematicae 31 (2020), 791–808; [arXiv:1912.01162](#).